Keenan Eikenberry

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https://keenan-eikenberry.github.io/

Research Interests

- Neural architectures (theory, design, and applications)
- Applications of optimal transport to machine learning and statistical inference

Current Position

2023 – Present
- **Postdoctoral Research Associate**, Dartmouth College
  Department of Mathematics, Hanover, NH
  Supervisors: Yoonsang Lee and Anne Gelb

Education

2019 – 2023
- **Ph.D. in Applied Mathematics**, Arizona State University
  Advisors: Doug Cochran and Shiwei Lan
  Diss.: *Bayesian Inference for Markov Kernels Valued in Wasserstein Spaces*

2014 – 2016
- **M.A. in Mathematics**, Arizona State University
  Advisors: John Quigg and Steven Kaliszewski
  Thesis: *Functorial Results for C^*-algebras of Higher-Rank Graphs*

2009 – 2014
- **B.S. in Mathematics**, Arizona State University
  B.A. in English Literature (minor in Philosophy), Arizona State University
  Summa cum laude, GPA: 4.0

Papers and Code

Papers

1. **Eikenberry, K.**, Liu, L., & Lee, Y. (2026). Post-training augmentation invariance. *Transactions on Machine Learning Research*. Retrieved from https://openreview.net/forum?id=Z4uUwU6zRe
2. Eikenberry, S. E., Mancuso, M., Iboi, E., Phan, T., **Eikenberry, K.**, Kuang, Y., … Gumel, A. B. (2020). To mask or not to mask: Modeling the potential for face mask use by the general public to curtail the covid-19 pandemic. *Infectious disease modelling*, *5*, 293–308.

Code

1. Post-Training Augmentation Invariance. (2026). https://github.com/keenan-eikenberry/augmentation_invariance.

Selected Talks

- Wasserstein Dependence and Representation Learning. Applied and Computational Mathematics Seminar, Dartmouth College. November 2024.
- Maximizing Multiview Wasserstein Dependence in Autoencoders. SIAM Mathematics of Data Science Conference, Atlanta, Georgia. October 2024.

Selected Talks (continued)

- Markov Kernels Valued in Wasserstein Spaces. Online CodEx Seminar. July 2023.
- Stacks of C^*-Categories. Great Plains Operator Theory Symposium, Texas A&M University. May 2019.
- Group Actions on Topological k-Graphs. C*-Algebra Seminar, Arizona State University. November 2016.
- k-Graphs from Cayley Graphs. Great Plains Operator Theory Symposium, University of Illinois at Urbana-Champaign. May 2016.
- Functoriality for Higher-Rank Graph Algebras. C*-Algebra Seminar, Arizona State University. March 2016.

Skills

Mathematics	Analysis, optimal transport, probability, category theory, partial differential equations, differential geometry, algebraic topology, abstract algebra, operator algebras
ML and AI	Neural architectures, theory of deep learning, self-supervised representation learning, optimal transport-based methods, scientific machine learning
Programming	Python (expert in PyTorch), Julia